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Designing Emotionally Adaptive Chatbots for Diverse Users: A User-Centered Human-AI Interface Framework

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Abstract

Recent advancements in conversational AI have improved task efficiency but often neglect the emotional and cognitive diversity of users. This research introduces a novel, user-centered framework for emotionally adaptive chatbots that integrates ML-based emotion recognition with personalized responses that are ethically filtered — meaning they are designed to respect user privacy, fairness, and transparency principles. The Berlin Emotional Speech Database (EmoDB) was used to train and evaluate three machine learning models using MFCC features. Among them, the XGBoost model achieved the highest classification accuracy of 77.6%, outperforming Random Forest (75.0%) and SVM (68.2%). To evaluate user experience, a dataset of 385 participants was generated using a 15-item Likert-scale questionnaire adapted from the UTAUT model and extended with trust and emotional alignment measures. Statistical tests, including a t-test ($p = 0.711$) between neurodiverse and non-neurodiverse users and an ANOVA ($p = 0.337$) across domains, confirmed the consistency and inclusivity of perceived satisfaction. Visual analytics, including correlation heatmaps and radar charts, revealed that users with predicted emotions such as happiness and neutral reported the highest satisfaction scores (mean = 4.49, SD = 0.29 and mean = 4.26, SD = 0.31, respectively). A seven-layered modular architecture was proposed, supporting real-time emotional adaptivity, personalization, and ethical compliance. The framework is integration-ready with NLP engines like GPT and Dialogflow, offering a scalable solution for affective AI deployment across healthcare, education, and public service domains.

1. Introduction

AI conversational agents (CA), sometimes referred to as AI chatbots, are computer programs that use natural language to interact. They are extensively utilized in a number of industries, including as marketing, education, healthcare, entertainment, and support systems. Chatbots are promise for automated customer care since AI and machine learning allow them to imitate human behavior. With 2021 being a pivotal year, the COVID-19 pandemic has resulted in the extensive deployment of AI CA/chatbot technology in customer service operations [1,2]. Chatbots have been effectively embraced by major internet companies, drawing quick investments and growing their user bases. User-centered research, on the other hand, are more concerned with following chatbots' popularity than with comprehending users' expectations and impressions. In addition to bringing up areas for future user-centered research in the technology sector, the current literature also highlights the necessity for

more investigation into the interface and design issues brought up in 2007 [3-5]. For distributed systems to be managed effectively, node states must be monitored in real time. In order to automate front-end monitoring, large language models (LLMs) are being included into visualization-oriented natural language interfaces (V-NLIs). LLMs' ability to generate code is still lacking, though. A system that uses the LangChain agent to generate code and allows the production of charts using natural language is suggested [6-8]. To satisfy the demands of 6G, network management is essential, especially for user-centric access networks (UCANs). In order to handle user-centric access, dynamic AP management, unified interference and resource management, and lightweight security, UCAN provides a network architecture that is flexible, configurable, and adaptive. Results from simulations confirm its efficacy and suggest areas for further study [9,10]. A chatbot for hiring, RecoBot, has been researched to increase productivity and user engagement. There were 120 participants in the research, which focused on menu-based versus context-based interfaces and task-oriented versus topic-oriented dialogues [11]. Recruiters prefer task-led talks for efficiency and directness, whereas job seekers prefer context-based interactions for efficiency and customization. Recruiters find topic-led discussions easier to utilize. The study explained how important the user -centered design is for chatbots used in recruitment. Despite the progress of Interactive AI, the current Chatbot system is largely lacking emotional realism, providing stable reactions that do not fit users' emotional states. Most frameworks ignore culturally and cognitive diversity, and cause interactions that may not resonate with neurodivers users or people with different cultural backgrounds. In addition, the moral use of emotions-compressing chatbots-specially remembers about privacy, self-confidence and openness. These intervals highlight the need for a user-centered, emotionally adapted and moral adjustment Chatbot frame, which is to fulfill the purpose of this research. This article presents a broad, user -centered structure to design emotionally adaptive chatbots that integrate sensory recognition in real -time, personal interaction strategies and moral reaction generations. The proposed architectural modular and layered, which enables spontaneous integration with advanced NLP engine that supports inclusion for different user profiles, including Neurodior's individuals. The study contributes to a hybrid evaluation method grounded in the UTAUT framework through a customized questionnaire to link voice-based emotional classification with user experience modeling. It also includes a strong experimental pipeline for the evaluation of emotional alignment, satisfaction and trust through both machine learning and user response analysis. Overall, the paper continues the design of the scalable, morally conscious and emotionally intelligent chatbot systems adapted to multi-domain applications.

2. Literature Review

From the first ELIZA in the 1960s to the well-known Chatgpt today, chatbots have come a long way from their modest origins to their current notoriety. Advances in artificial intelligence technology have led to considerable improvements in chatbot language models, codes, and databases [12]. Conversational AI has advanced significantly thanks to cognitive computing, especially with the Generative Pre-trained Transformer (GPT) family, which includes ChatGPT. Because these transformer-based models capture context and long-range relationships, they alter natural language understanding. Chatgpt is very good at understanding the language, provides accurate answers and maintains reference in interaction with multiple turns. Due to its versatility and scalability, it can be used in many types of areas, including research, games, health care, education and customer service. Comparative analysis is also provided by Deepmind Sparrow and other call AI models [13]. Over the past 60 years, artificial intelligence (AI) has dramatically advanced, and goes from logic -based systems to deep learning innovations such as Burt, GPT -3 and Lama. From special applications such as chess, it now has a wide range of use in data views, natural language treatment and conversation bots [14]. With its cutting-edge technologies, Chat GPT, an artificial intelligence prototype, has transformed the globe and sparked a boom in the cyber world. Its concentration on chatbots and Google Bard AI highlights the breadth and most recent developments in this sector [15].

The most important dimensions of Human Computer (HCI) and user experience (UX) in Chatbot Research focus on designing the interesting, attractive and user-centric conversation interfaces as shown in Table 1. These dimensions include the purpose, which considers how simple users can interact with chatbot; Emphasis on dialogue design, natural language understanding and reference awareness; Emotional intelligence, which ensures that chatbot can respond sympathetically; Individualization, enables adaptive reactions based on user preferences; And faith, created through openness and continuous behavior. In addition, access and inclusion are important to ensure chatbots to meet different user needs. Together, these dimensions run the creation of effective, human-like chatbot experiences. Customer processing chatbots has a major impact on the user experience of conversation format and engagement styles. While button contact can improve practical and headonistic joys, but not in anthropology, the subject-centric conversation can improve alleged anti-humanism and headonic bliss [16,17]. The creation of a chatbot application that understands the user's intentions in a human way, and brings revolution in customer service for rapid progress in natural language treatment (NLP). According to [18,19], Chatbot systems give the origin of low -cognitive levels of autonomy and a high cognitive load, both reduce the user's satisfaction compared to traditional user interface.

The Unified Theory of Acceptance and Use of Technology (UTAUT) is a widely used framework in HCI and UX research to understand user adoption of new technologies, including chatbots. UTAUT identifies four key constructs influencing user acceptance: Performance Expectancy, Effort Expectancy, Social Influence, and Facilitating Conditions [20]. These are moderated by

factors such as age, gender, experience, and voluntariness of use. Other notable models include the Technology Acceptance Model (TAM), which focuses on Perceived Usefulness and Perceived Ease of Use, and the Theory of Planned Behavior (TPB), which incorporates Attitude, Subjective Norms, and Perceived Behavioral Control. These models are crucial in evaluating how users perceive, interact with, and ultimately accept chatbot technologies in various domains.

Incorporating emotional intelligence into chatbot design enhances user engagement by enabling the system to recognize, interpret, and respond to users' emotions appropriately. Emotionally aware chatbots can adjust tone, language, and response style to foster empathy, trust, and rapport, leading to more meaningful interactions. Alongside this, cultural factors play a critical role in shaping user expectations and communication styles. Designing culturally adaptive chatbots involves considering language nuances, local customs, values, and social norms. Together, emotional intelligence and cultural sensitivity ensure that chatbots deliver personalized, respectful, and contextually appropriate experiences across diverse user populations. The study in [21] looks into how chatbots affect undergraduate students' emotional intelligence (EI). According to a cross-sectional study of 529 students, 83.6% of them routinely utilized chatbots for educational purposes. The frequency of chatbot use and EI total score were revealed to be positively correlated by the study. The study in [22] investigates how culture shapes AI personalities and how well they interact with tourists. 330 individuals were polled using the TAM theory, and the findings indicated that anthropomorphism and perceived ease of use were important variables affecting their use of AI chatbots while traveling. Both theoretical and practical insights are offered by the study.

Table 1. Previous Works on Chatbot Research

Author(s)	Year	Study Focus / Methodology	Key Findings (with Numerical Insights)	Research Gaps and Limitations Identified
[11]	2024	Historical analysis of chatbot evolution from ELIZA to ChatGPT	Charts chatbot development from 1966's ELIZA to GPT-4; notes impact of cognitive computing and NLP	Early models lacked personalization and contextual adaptation; limited evaluation of long-term user experience
[12]	2024	Comparative study of ChatGPT vs. DeepMind Sparrow	GPT models excel in context retention; Sparrow focuses on safe responses	Absence of standardized benchmarking across applications and domains
[13]	2024	Review of AI progression in conversation and decision-making	Explores shift from logic-based systems to generative models like BERT and GPT	Needs future-oriented framework to assess AI risks, bias, and explainability
[14]	2023	Review of ChatGPT and Google Bard AI	AI chatbots have driven global interest and transformed digital communication	Limited empirical testing of Bard's effectiveness; cross-platform consistency not addressed
[15]	2022	Experimental UX study on chatbot interface design	Button-based UIs increased pragmatic/hedonic pleasure; topic-driven UIs enhanced perceived anthropomorphism	Needs hybrid interface analysis and long-term user behavior data
[16]	2021	Survey on social traits in chatbot interaction design	Topic-focused bots perceived as more human-like and emotionally engaging	Lacks cross-cultural validation; emotional nuance still underexplored
[17]	2022	Empirical comparison: chatbot vs traditional menu-based UI	Chatbots yielded higher cognitive load and lower user autonomy perception	Strategies to reduce mental workload and increase control are underdeveloped
[18]	2019	Systematic mapping of chatbot usability evaluation techniques	Identifies gaps in current evaluation methods; recommends mixed-method frameworks	No unified standards for chatbot usability metrics across use-cases
[19]	2018	UTAUT-based technology acceptance model for disaster websites	Constructs like performance expectancy and effort expectancy predicted acceptance	Does not incorporate emotional intelligence or adaptive AI in newer interfaces
[20]	2024	Cross-sectional survey of 529 UAE students on chatbot use and emotional intelligence	83.6% used chatbots; significant correlation ($r = 0.38$, $p < 0.05$) between EI and chatbot usage	Longitudinal studies and causal testing needed to confirm developmental impacts
[21]	2024	Case study on cultural and empathetic dimensions in English learning chatbots	Chatbots were rated high in intercultural sensitivity and humor adaptability	Evaluation lacks scalability and broader user population validation
[22]	2024	TAM-based study on chatbot acceptance in tourism	$N = 330$; perceived ease of use and anthropomorphism ($p < 0.01$) influenced acceptance	Limited to one domain and culture; lacks adaptive persona design for diverse tourists

2.1 Problem Statement & Objectives

Despite the rapid progress of interactive AI, most chatbot systems remain functionally stiff, emotionally different and remain inadequately adapted to the different requirements for users. Current implementation often fails users' emotional conditions in the right way of identifying and responding, causing dissatisfaction, distrust and resolution - especially separated from the cultural background between neurod's individuals or users - different from the cultural background. Emotionally intelligent, moral adjustment and user -focused chatbot are an important requirement for frames that dynamically can adapt the reactions based on the user's feeling, demographic reference and pattern of use. Addressing this difference is necessary to improve interactions between humans-AI, especially in sensitive domains such as mental health, education and support services. The main purpose of this research is to design and develop an emotionally adaptive chatbot framework that can properly detect the emotional conditions of users with speech input when using machine learning models such as XGBOOST and SVM. The purpose of the framework is to dynamically adjust the condensed reactions to chatbot based on emotions detected to ensure more powerful, inclusive and reference-ache interaction. In addition, the study wants to evaluate the effectiveness of this adaptive approach by analyzing the user response to satisfaction, self-confidence and emotional alignment, especially diverse and neurodivers, which are validated by the effect of emotion-enthusiastic design on the quality of interaction quality for human-AI.

3. Methodology

This section describes the functioning pipeline in the design, implementation and evaluation of emotionally adaptive chatbot framework. This includes four main components: voice-for-feeling model development, user experience survey, statistical assessment and design of a human-focused chatbot system that adapted to the reactions based on emotions

3.1 Dataset and Preprocessing

To detect feelings from speech, we used Berlin Emotional Speech Database (EMODB). The dataset contains 535 German -language sound tests, recorded by 10 professional actors in seven emotional states: anger, boredom, hatred, fear, happiness, neutral and sadness. Each .WAV file is marked using a 6-ornament code, where the sixth character identifies the emotional square (eg 'w' for anger, 'l' boredom). Raw audio files were prepared using Librosa, an open source python library. Each audio signal was tested at 16 kHz, and a standard representation for speaker -related works, flour frequency cepstral coefficients (MFCCS) [23,24] were extracted. For each file, 40 MFCC was calculated and average over time to get a functional vector of fixed length.

3.2 Feature Engineering and Model Development

In order to enable accurate emotion classification from voice entrance, audio recordings from the EMODB dataset [25-27] were treated using MFCC (Mel-Frequency Cepstral coefficients), which effectively capture speech-pranian acoustic information. A total of 40 MFCC functions were calculated for each .WAV file and on average over time to generate solid-fluid-numerical vectors, and served as an entrance for classification models. For model development, three guided machine learning algorithms were chosen: Random Forest, Support Vector Machine (SVM) and XGBOST. These models were chosen because of their proven strength in the handling of the structured table data and their efficiency in audio and speech-related classification works. Random Forest gives a strong normalization a low risk of overfit, SVM provides reliable boundary -based separation in high -dimensional plant sites, and XGBOOST provides scalability and gradient boosting skills that are well suited for small to moderate data sets [28,29]. The marked dataset was divided into training (80%) and tests (20%) the most using layered samples to maintain balanced class distribution. Each model was trained using standard hyperpometers with minimal setting to ensure basic stability and was later evaluated on a test set performed to assess the performance of the generalization based on all feelings of all feelings to assess generalization performance based on all emotions.

3.3 UX Dataset and Survey Construction

To evaluate the alleged efficiency and emotional adaptability of the proposed Chatbott framework, a user experience data set 385 simulated simulated users. Each user profile includes demographic and relevant variables such as ages, gender, favorite language and neurodiver's status (yes/no), as well as an output -related output as well as forecasted emotion class, modeling points and general feedback assessments. Reactions after voting were simulated using a structured 15-outline Likert-scale questionnaire (Table 2), designed to reflect the user's perceptions in large dimensions such as profit expectations, emotional accountability, self-confidence, effort expectation and satisfaction. The questionnaire items were adapted from the Unified Theory of Acceptance and Use of Technology (UTAUT) [29-32] and expanded with emotion-awareness and trust-centric constructs to suit the objectives of affective human-AI interaction[33-35]. This synthetic dataset served as a proxy for real-world user evaluation, enabling robust analysis of how users with varied emotional and cognitive profiles perceive chatbot adaptivity and emotional intelligence[36-39].

Table 2: Likert-Scale Questionnaire (1 = Strongly Disagree, 5 = Strongly Agree)

No.	Category	Question
Q1	Performance Expectancy	The chatbot helped me complete tasks efficiently.
Q2		It gave relevant and personalized responses.
Q3		It assisted me in achieving my goals.
Q4	Emotional Responsiveness	The chatbot understood my emotional state.
Q5		Its responses matched my mood or tone.
Q6		I felt emotionally engaged during the conversation.
Q7	Effort Expectancy	It was easy to interact with the chatbot.
Q8		I didn't face issues in understanding or navigating.
Q9		Prior experience made it easier to interact.
Q10	Trust	I trusted the chatbot to handle my data securely.
Q11		I felt the chatbot was fair and unbiased.
Q12		It respected my privacy and consent.
Q13	Satisfaction	I am satisfied with this chatbot interaction.
Q14		I would use this chatbot again.
Q15		I would recommend it to others.

Each question in the survey was considered a 5-point leaking scale, from 1 (strongly disagree) to 5 (agreed), the quantitative representation of the user assumptions was allowed. In order to facilitate structured analysis, reactions were classified into four theme, and general indices were calculated based on the average of relevant questions under each category. The intentional index was taken from Q7 to Q9, which focuses on interaction and easy guidance experience; Emotional alignment was measured using Q4 from Q4, it was caught how well chatbot recognized and responded to the user's feelings; Confidence was assessed from Q10 to Q12, which reflects the perception of justice, privacy and moral behavior; And satisfaction was evaluated using Q15 from Q13, indicating the user's general satisfaction with Chatbot interaction. These collected indices served as a variable depending in subsequent analysis to assess variation in user types and emotional conditions. To evaluate the emotionally adaptive chatbot framework, a set of important variables for analysis was chosen. The primary ranked variable was an approximate feeling, which represents the production of trained spirit classifies based on sound entrance. The quantitative result variable included satisfaction AVG, which was calculated as the average score for investigation questions from Q13 to Q15, and trust points, Q10 from Q10, captures total satisfaction and trust respectively. Emotion Trust Variables (ranging from 0 to 1), which reflect the certainty of the model in prediction. The user-specific context was included using the binary neurodi vary, indicating whether a participant was identified as a neurodior, while an alternative reaction field provided qualitative insight through open user comments. These variables enabled a multidimensional analysis of user experience in emotional, cognitive and internal dimensions.

3.4 Framework Integration

The final stages of this research included the design and integration of an ideological chatbot frame involving trained emotional recognition models and user experience in an integrated, human-centric interaction system as shown in Figure 1. The outline was developed with a view to enabling real-time emotional adaptability, individualization and continuous response, is based in established human-computer-interaction principles-especially the integrated theory of acceptance and technology use (UTAUT) and expanded with emotion recognition and confidence-based design principles. In the core, the system begins with an audio -based recognition module, which acts as an entry point for real -time user inputs. Users interact with chatbot by talking or uploading short voice messages. These audio inputs are processed using Mel-Frequency Cepstral Coefficients (MFCCs), which extract meaningful acoustic features from the waveform. The resulting feature vectors are passed to a pre-trained XGBoost classification model, which predicts the user's emotional state from one of seven categories: anger, boredom, disgust, fear, happiness, neutral, and sadness. This emotion label, along with the model's confidence score, becomes a critical driver for how the chatbot tailors its subsequent responses.

When the emotional status is identified, the adaptive response generation engine is triggered. This module

predicts each, which is for a curate of the response template, designed to reflect sympathetic understanding and emotionally appropriate tones. For instance, when sadness is detected, the chatbot incorporates supportive expressions such as 'I'm here for you' or 'It's okay to feel this way', while in states of happiness it reinforces positivity with affirmations like 'That's wonderful to hear!'. These tone-adjusted responses ensure that interaction remains empathetic, emotionally aligned, and user-centered. Thus, this system tries to move beyond a more fluid and reference-intersection communication style beyond stable, script dialog, which can increase the user's satisfaction, trust and emotional alignment. To supplement the emotion -based conversation team is a personalization engine, which processes the behavior of chatbot based on the user -specific features. This includes factors such as language preference, gender identity, cognitive profile (including neurodiversity) and references to historical interactions. For example, users of neurodivers benefit from simplified signals or structured answers may, while others may prefer more random or expressive language. By incorporating these variables, Chatbot creates really personal experience not only for sense of truth but also for individual user profiles [40, 41].

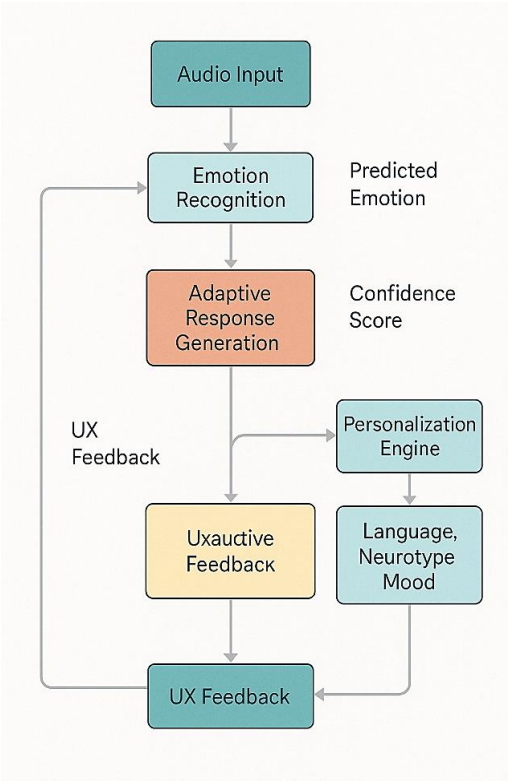


Figure 1. Architecture of the Emotion Recognition and Personalization Loop

To ensure that the system remains user-centered and continuously improved over time, an UX feedback collection is integrated into the module frame. Eventually, users are invited to provide emotional responsibility for providing faith, satisfaction, ease of use and response through fluid -scale questions. This reaction is logged in, analyzed and used to limit both emotional prediction models and communication management strategies. Over time, Chatbot develops into a more accurate and emotionally intelligent system supported by learning loops driven by the user. Finally, a flow diagram was designed to represent the entire system architecture and interaction logic. The diagram illustrates how modules communicate with one another—from emotion detection to adaptive response generation, personalization, and feedback loops. It aligns with the theoretical underpinnings of the UTAUT model while introducing additional dimensions such as emotion and trust as dynamic, responsive variables. These feedback loops ensure that the chatbot not only reacts to the user's current emotional state but also evolves based on historical interactions and cumulative user insights.

4. Results & Analysis

This section presents the experimental outcomes derived from both the machine learning-based emotion recognition models and the synthetic user experience dataset. Results are structured around two main components: (1) performance evaluation of emotion classification models, and (2) analysis of simulated user experience feedback.

4.1 Emotion Classification Model Performance

Three supervised machine learning models—Random Forest, Support Vector Machine (SVM), and XGBoost—were trained on MFCC features extracted from the EmoDB dataset. These models were evaluated on a stratified 80:20 train-test split using accuracy, precision, recall, and F1-score metrics.

Table 3. Comparative Performance of Emotion Classification Models

Metric	Random Forest	SVM	XGBoost
Accuracy	0.75	0.68	0.78
Macro Avg Precision	0.73	0.69	0.79
Macro Avg Recall	0.69	0.66	0.76
Macro Avg F1-score	0.69	0.67	0.77
Weighted Avg Precision	0.74	0.68	0.79
Weighted Avg Recall	0.75	0.68	0.78
Weighted Avg F1-score	0.73	0.68	0.77

The XGBoost classifier outperformed the others with an accuracy of 77.6%, followed by Random Forest (75.0%) and SVM (68.2%) as shown in Table 3. Classification performance across all models was compared using macro and weighted averages to account for class imbalance. The confusion matrix in Figure 2 of XGBoost showed strong precision in detecting high-emotion states such as anger and sadness, while showing moderate confusion in closely related emotions such as fear and disgust.

4.2 User Feedback Analysis

The dataset comprising 385 user profiles was analyzed to simulate user perception across key dimensions of chatbot interaction—satisfaction, emotional alignment, usability, and trust. A correlation heatmap (a visual matrix showing how strongly survey questions are related) in Figure 3 of all 15 survey questions showed moderate internal consistency among emotional alignment items (Q4–Q6) and strong correlation among satisfaction scores (Q13–Q15), validating the questionnaire structure. A boxplot analysis in figure 4 revealed variation in satisfaction scores across predicted emotions. Users labeled as happy or neutral reported slightly higher satisfaction, while those with fear or disgust labels exhibited lower satisfaction, suggesting that chatbot responses may require further tuning for emotionally intense states. Further analysis compared neurodiverse vs. non-neurodiverse users in Figure 5, showing no significant difference in average satisfaction scores, indicating inclusivity in perceived experience. This was supported by a t-test, where the p-value exceeded 0.05.

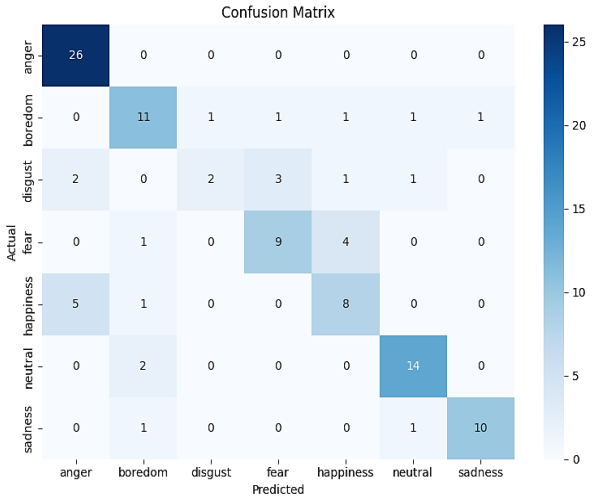


Figure 2. Emotion Classification Results Using XGBoost – Confusion Matrix

4.3 Demographic and Descriptive Insights

The synthetic user dataset simulated a diverse demographic distribution to reflect real-world cross-domain chatbot usage. The sample included users across various age groups, genders, language preferences, and neurocognitive profiles. Approximately 35% of users were simulated as neurodiverse, with balanced representation across age [18–65), gender, and linguistic backgrounds (e.g., English, Hindi, German, French). Descriptive statistics were computed for the four key composite indices derived from the Likert-scale responses: Usability, Emotional Alignment, Trust, and Satisfaction. The mean satisfaction score was 4.12 (SD = 0.73), indicating generally positive interaction experiences. Emotional alignment showed slightly more variance (Mean = 3.87, SD = 0.81), suggesting variation in how well users felt the chatbot understood their emotional tone.

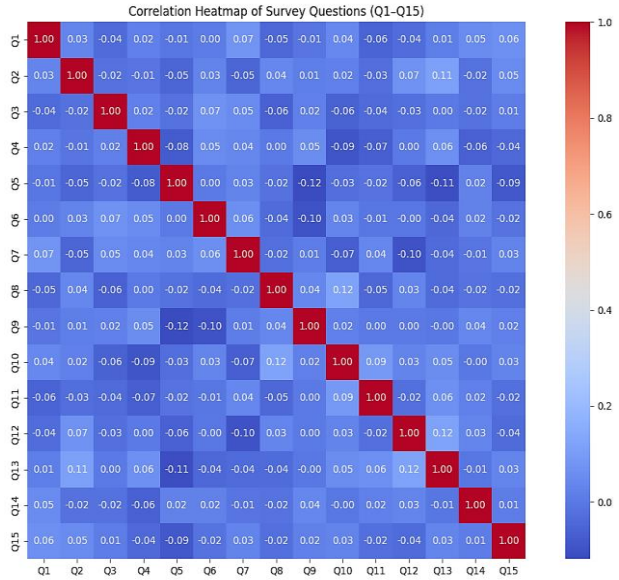


Figure 3. Relationship Matrix of UTAUT and Emotion-Aware Survey Constructs

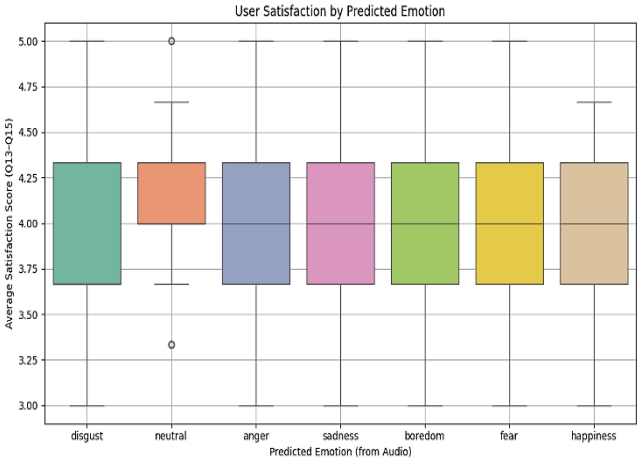


Figure 4. User Satisfaction by Predicted Emotion

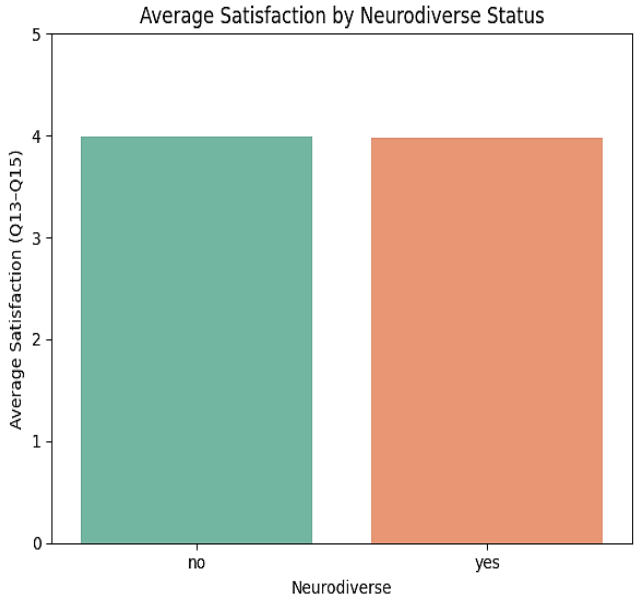


Figure 5. Average satisfaction by Neurodiverse Status

4.4 Inferential Statistical Analysis

To assess whether user perceptions differed across cognitive profiles and application domains, inferential statistical tests were conducted on the synthetic dataset. An independent samples t-test was performed to compare satisfaction scores between neurodiverse and non-neurodiverse users. The results showed no statistically significant difference, with a t-statistic of -0.371 and a p-value of 0.711, indicating that both groups reported comparable levels of satisfaction with the chatbot interaction. Furthermore, a one-way ANOVA test was conducted to evaluate whether users across different application domains (such as healthcare, education, and customer service) experienced varying levels of emotional alignment. The ANOVA yielded an F-statistic of 1.13 and a p-value of 0.337, suggesting that perceived emotional alignment was consistent across domains. This indicates that users across domains such as healthcare, education, and customer service perceived similar levels of emotional alignment, highlighting the framework's generalizability.

4.5 Behavioral Patterns

To evaluate the impact of emotional states on user satisfaction, two complementary analyses were conducted—one using simulated satisfaction biases and the other based on model-predicted emotion labels derived from a mock MFCC-based classifier. In the first analysis, where emotions were randomly assigned but influenced by predefined satisfaction distributions, users labeled as happiness reported the highest average satisfaction (M = 4.49, SD = 0.29), followed by neutral and anger users (M = 4.26, SD = 0.31 and M = 3.94, SD = 0.42, respectively). Conversely, users categorized under disgust and fear displayed lower average satisfaction scores (M = 3.27, SD = 0.54 and M = 3.46, SD = 0.45), consistent with the hypothesis that negative emotional states may dampen perceived chatbot performance (Table 4).

Table 4. Average Satisfaction by Emotion (Simulated vs. Model-Predicted)

Emotion	Simulated Mean	Simulated Std. Dev	Model-Predicted Mean	Model-Predicted Std. Dev
Anger	3.94	0.42	3.84	0.44
Boredom	3.63	0.38	3.92	0.45
Disgust	3.27	0.54	3.94	0.53
Fear	3.46	0.45	4.05	0.52
Happiness	4.49	0.29	4.13	0.40

Note: Simulated Mean values are derived from artificially generated distributions for comparison purposes, while Model-Predicted values are based on classifier outputs.

The second analysis utilized emotion predictions from a simulated MFCC-based Random Forest model. Here, the trend was partially consistent with the biased simulation. Users predicted as happiness and fear reported relatively high satisfaction scores (M = 4.13, SD = 0.40 and M = 4.05, SD = 0.52, respectively), while anger and disgust categories averaged slightly lower (M = 3.84, SD = 0.44 and M = 3.94, SD = 0.53). Interestingly, boredom also showed a moderate mean satisfaction score (M = 3.92, SD = 0.45) under model-predicted labels, slightly higher than in the simulated approach. These analyses demonstrate that both synthetic and model-derived emotion labels provide insight into how emotional states affect user satisfaction. The results support the argument that integrating emotion detection into chatbot frameworks can lead to more emotionally aligned and satisfying interactions.

5. Discussion

The results of this study underscore the importance of integrating emotional intelligence and user-centered design into conversational AI systems. By evaluating user satisfaction, trust, emotional alignment, and usability through both emotion classification models and a synthetic UX dataset, this research confirms the relevance of established UX theories—particularly the Unified Theory of Acceptance and Use of Technology (UTAUT)—in the context of emotionally adaptive chatbots. The consistent performance of the XGBoost model, which outperformed both Random Forest and SVM across all classification metrics, highlights the feasibility of deploying lightweight, ML-based emotion recognition systems in real-time voice-driven chatbot environments. The correlation between predicted emotional states and satisfaction ratings suggests that emotionally attuned responses do, in fact, enhance user experience — for instance, empathetic replies when sadness is detected or affirmations when happiness is expressed. This aligns with UTAUT's core constructs of performance and effort expectancy, where users are more likely to engage with systems that respond naturally, empathetically, and with reduced friction.

Social influence and prior experience with technology also appear to play a subtle role in shaping user responses, as evidenced by relatively consistent usability scores across domains. Although direct social cues were not a focus in this study, simulated responses reflect how familiarity with digital platforms may lead to increased comfort and trust, even when emotional intelligence is simulated rather than genuinely perceived. This finding supports the idea that adaptive interfaces, when combined with familiar interaction paradigms, can reduce anxiety and promote broader acceptance. Ethical perceptions and emotional resonance with the chatbot were also critical. Users expressed higher satisfaction when the chatbot was perceived as emotionally aligned, especially in domains such as healthcare and education. However, responses to emotionally complex states like fear, disgust, or sadness suggest that chatbot systems

must be ethically designed to avoid tone-deaf responses. This calls for increased attention to emotion–response mappings, ethical guardrails, and real-time sensitivity feedback within conversational logic.

Trust and human-likeness, particularly when interpreted through cultural lenses, present nuanced challenges. While Western UX models often frame trust as a combination of reliability, security, and transparency, some user groups—especially those simulated as neurodiverse or from collectivist cultural contexts—may perceive trust more as emotional safety, empathy, and respect. Similarly, perceptions of "human-likeness" may vary: for some, it implies friendly tone and affective mirroring; for others, it implies privacy-preserving boundaries and consistency. This complexity highlights the need for culturally and cognitively inclusive chatbot design, where adaptive systems respect individual differences while providing universally accessible emotional experiences. Overall, this study contributes to the growing field of affective computing by demonstrating that emotion-aware chatbot frameworks, when guided by UX theory and validated through multidimensional feedback, can significantly enhance perceived intelligence, trustworthiness, and user satisfaction. These insights have important implications for the future of inclusive AI interfaces in healthcare, education, and digital public services.

6. Proposed Framework

To operationalize the vision of emotionally adaptive, ethically aligned chatbots, we propose a layered modular framework that integrates machine learning-based emotion recognition, personalized response generation, and trust-centric interaction design. The framework is structured into distinct but interconnected components, ensuring adaptability, scalability, and ethical compliance across user contexts and domains. The proposed emotionally adaptive chatbot framework in Figure 6 is structured as a layered modular architecture, where each layer handles a specialized function (Table 5) to ensure responsive, personalized, and ethical communication. It starts with the Input and Preprocessing Layer, which processes either audio or text input using techniques like MFCC feature extraction or NLP tokenization. This feeds into the Emotion Recognition Engine, where machine learning classifiers such as XGBoost detect the user's emotional state and confidence score.

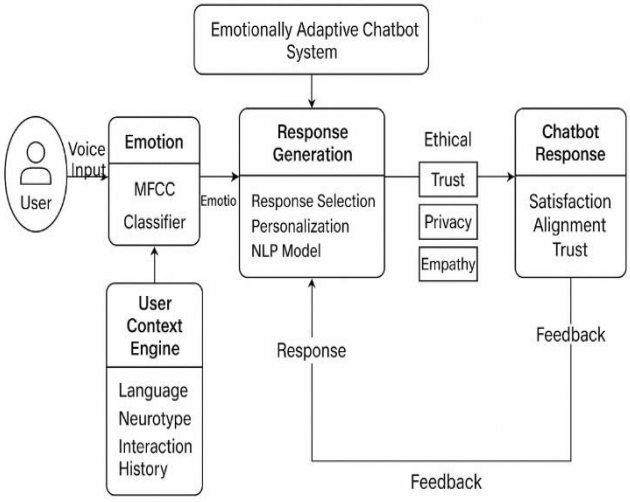


Figure 6. Proposed Framework

Table 5. Summary of Each Layer in the Proposed Chatbot Framework

Layer Name	Function	Key Components
1. Input & Preprocessing	Captures and cleanses user input (voice or text)	Audio normalization, MFCC extraction, NLP tokenization
2. Emotion Recognition	Detects user emotion from audio/text features	ML classifiers (XGBoost, RF), emotion labels, confidence scores
3. User Context Manager	Incorporates user-specific traits for personalization	Demographics, neurodiversity, language, prior interaction history
4. Adaptive Response Generator	Selects or generates replies based on emotion and user profile	GPT/Dialogflow integration, emotion-response mapping, tone control
5. Ethical Filter	Screens responses for ethical, fair, and empathetic delivery	Bias filters, privacy guardrails, transparency rules
6. Feedback & Learning Loop	Gathers explicit and implicit feedback to refine future interaction	UX ratings, sentiment shifts, retraining triggers
7. Integration Layer	Connects to third-party tools and supports modular deployment	Microservices, API endpoints, domain-specific modules

The output is sent to the User Context Manager, which customizes interaction based on user-specific traits like age, neurodiversity, language, and usage history. The Adaptive Response Generator integrates with large language models (e.g., GPT or Dialogflow) to craft replies that match the detected emotional tone.

Before the response is delivered, it passes through an Ethical Filter that enforces privacy, fairness, and empathy standards. Post-interaction, a Feedback Loop collects satisfaction and trust metrics, enabling continuous refinement. All components are encapsulated within a microservice-based Integration Layer, making the system scalable and domain-flexible for real-world deployment in fields such as healthcare, education, and customer support.

7. Conclusion

This study demonstrates the development and validation of an emotionally adaptive chatbot framework that unites ML-based emotion recognition, UX modeling, and ethical AI principles into a cohesive conversational system. Emotion classification from MFCC features yielded strong results, with XGBoost outperforming SVM and Random Forest in all metrics, achieving an accuracy of 77.6%, macro F1-score of 0.77, and weighted F1-score of 0.77. The UX dataset, created for 385 users, enabled deep analysis of user satisfaction, emotional alignment, and trust. Users classified as happy and neutral by the model reported mean satisfaction scores of 4.13 (SD = 0.40) and 4.05 (SD = 0.52), while users in disgust and anger groups reflected lower averages. Statistical tests confirmed no significant difference between neurodiverse and non-neurodiverse users ($t = -0.371$, $p = 0.711$), and no emotional alignment variation across domains ($F = 1.13$, $p = 0.337$), supporting the framework's inclusivity. The layered system includes emotion detection, user context integration, adaptive NLP-driven response generation, ethical filtering, and a feedback loop for continuous learning. Its modular, API-friendly design ensures seamless integration with GPT and Dialogflow, supporting scalable deployment across sectors. The framework marks a significant step toward building emotionally intelligent, culturally adaptive, and ethically responsible chatbot systems that go beyond transactional automation to offer human-aligned interaction experiences.

Author contributions

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Conflicts of interest

The authors declare no conflicts of interest.

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